

Executive Summary

Predicting Stock Market Trends Using Financial News Sentiment

Client: Alpine Capital Management AG

Timeline: 22 weeks

Alpine Capital manages >1B CHF in assets and operates a quantitative trading desk that processes thousands of transactions daily. The firm's current models for stock price direction prediction rely heavily on historical price data, missing critical insights from news, analyst commentary, and social media. Competitors using NLP-enhanced models consistently outperform Alpine's strategies by a significant percentage.

We have been tasked with building a multi-modal machine learning pipeline that integrates financial time-series data with news sentiment analysis to improve daily stock price direction predictions. We need a technical team to design, implement, and evaluate models while ensuring interpretability, risk awareness, and regulatory compliance readiness.

Business Challenge

- **Information Gap:** Traditional models miss 40% of market-moving events reflected in news and social media
- **Diminishing Returns:** Purely technical strategies show declining returns as markets become more efficient
- **Regulatory Pressure:** [MiFID II](#) requires explainable algorithmic trading decisions; non-compliance can lead to significant penalties
- **Competition:** Hedge funds using NLP-enhanced models consistently achieve superior returns

Objectives

1. **Enhance Prediction Accuracy:** Improve daily direction prediction accuracy beyond baseline ARIMA and time-series models
2. **Integrate Alternative Data:** Incorporate financial news sentiment into predictive features
3. **Maintain Interpretability:** Clear attribution of predictions to price vs. sentiment factors
4. **Demonstrate Risk-Aware Trading:** Simulate risk scenarios to ensure predictions are robust under volatile conditions

Data Assets:

- 10 years of S&P 500 historical prices
- Finance-related news pulled from API

- Dataset of financial news sentences annotated for sentiment (positive, negative, or neutral) by domain experts

Regulatory & Compliance Requirements

MiFID II Article 17 + RTS 6 (EU Markets):

If your system can place or shape orders, you must build it so it behaves safely, is tested before use, is monitored while live, and leaves a trail that lets supervisors understand what happened.

- **Design & Data Preparation:** Firms must have effective systems and risk controls “suitable to the business,” including limits that prevent disorderly trading and erroneous orders. If parts of the system are outsourced or procured, the firm remains responsible and must keep the documentation needed to control them
- **Model Development & Pre-deployment Testing:** Before any first use or major change, the firm must have a clear method to design and test algorithms; a named senior manager must authorize deployment. Testing must show the algorithm does not behave in unintended ways, follows trading-venue rules, and can be switched off if needed.
- **Backtesting, Simulation & Stress:** Firms must run stress tests at least annually to show the system keeps working in stressed market conditions. Supervisors expect scenario volumes and conditions to be clearly specified

FINMA Guidance 08/2024 + Circular 2023/1 (Swiss Requirements):

Swiss supervisors expect clear accountability for AI, sensible explanations for outputs, and lifecycle control over changes and third-party tools. The platform that runs the model must be reliable: controlled releases, strong logging, alerts, and a way to recover from failures.

- **Explainability & Communication:** Outputs should be understandable to decision-makers; limits and uncertainties should be stated so people do not over-trust the system.
- **Model Lifecycle & Change Control:** Changes to models (re-training, new data sources, new features) should be planned, documented, and reviewed; third-party models/tools should be assessed and monitored.
- **Monitoring & Incident Handling:** Ongoing monitoring for errors or undesirable effects and remediation plans are expected.
- **Engineering & Release Discipline:** Banks must manage operational risks with controls around development, deployment, and change; separate testing from production.

Alpine Capital Management AG Company Policy

- **Responsible Trading Policy:** Alpine Capital Management AG has the following company policy: “Traders must understand the underlying rationale and key drivers behind automated trading signals before acting on them.” Any tools developed for the company must enable compliance with this policy.

Stakeholders and Their Priorities

Stakeholder	Primary Concerns	Success Metrics
Chief Investment Officer	Returns, risk management	Sharpe ratio, maximum drawdown, profit factor
Head of Quant Trading	Model performance, latency	Prediction accuracy, execution speed
Portfolio Managers	Interpretability, integration	Prediction accuracy, ease of use
Compliance Officer	Regulatory adherence	Audit readiness, documentation
Trading Desk	Actionable signals	Returns

Project Implementation

Solution Components

Through this project, you will gain experience in the steps needed to implement an AI-backed algorithmic trading system. Concretely, you will implement the following pipeline components:

Financial Time-Series Models

- Process historical S&P 500 data with proper handling of missing trading days
- Engineer technical indicators used ubiquitously across the trading industry, including moving averages and RSI
- Implement time-series models from traditional approaches (ARIMA, GARCH) to modern deep learning (LSTMs)

News Sentiment Analysis Models

- Process financial news from multiple sources (Financial Phrasebank, StockNewsAPI) with different text formats and quality levels
- Implement sentiment extraction using domain-specific embeddings (FinBERT) and pre-trained models

Multi-Modal Integration Framework

- Handle temporal alignment between fluctuating news flow and market trading hours
- Develop modular approaches that combine separate price and sentiment models using meta-learning
- Implement end-to-end architectures that jointly process market data and text for unified predictions
- Create ensemble methods with performance-based weighting across different model types

Trading Strategy Evaluation System

- Implement realistic backtesting
- Calculate risk-adjusted performance metrics including Sharpe ratio, maximum drawdown, and Calmar ratio
- Create performance attribution analysis showing which components drive returns

Project-Specific Grading Details

Technical Requirements

Pipeline Assessment Criteria (What instructors will verify):

The final pipeline must demonstrate the following capabilities when tested with new market data:

- **End-to-end processing:** Accept market prices and news text, output directional predictions
- **Multi-source integration:** Successfully align and process data from different temporal frequencies (news and stock cycles)
- **Sentiment extraction:** Functional NLP pipeline producing sentiment scores
- **Model diversity:** At least 3 different approaches (time-series, ML, deep learning) implemented
- **Risk controls:**
- **Strategy simulation:** Realistic backtesting framework with proper performance metrics

Example Advanced Capabilities (For extension credit):

- Adversarial robustness to market regime changes

Report Requirements

Aside from the general requirements described in the Grading Rubric, for this project, it is expected that your report contain the following information:

Financial Performance Analysis

- **Sentiment Value-Add:** Quantitative comparison showing whether adding news sentiment improves evaluation metrics and by how much
- **Market Regime Analysis:** Performance breakdown during different market conditions (bull/bear/volatile)

Model Robustness Assessment

- **Temporal Stability:** Performance consistency across different time periods
- **Sentiment Ablation Studies:** Impact of removing news signals on prediction accuracy

Regulatory Compliance Assessment

- **MiFID II Algorithm Trading:** Document your testing approach and risk controls (e.g., maximum position limits, loss thresholds)
- **Explainability for Compliance:** Show how you can explain which factors (price vs. sentiment) drove a specific trading decision